

JAN KWONG

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EDUCATION

University of California, San Diego
M.S. in Computer Science (Anticipated)

Sep 2025 – Dec 2026 (GPA 3.8)

B.S. in Computer Science

Sep 2021 – Jun 2025 (GPA 3.8)

EXPERIENCE

UC San Diego

La Jolla, CA

Apr 2024 – Jun 2025

Computer Science Tutor

- Mentored 400 students enrolled in courses "Theory of Computation" and "Advanced Data Structures" during office hours, offering detailed feedback on assignments and exams to aid their understanding
- Participated in weekly meetings with a teaching team of 10 to discuss course progress and address challenges

UC San Diego LASR Lab

La Jolla, CA

Jan 2024 – Dec 2024

Undergraduate Research Assistant

- Co-author on a research paper submitted to the Acoustical Society of America and the PsyArXiv evaluating the performance of OpenAI's Whisper against 75 human transcribers on a dataset of 300 English sentences spoken by 20 speakers with varying accents
 - Developed Python scripts to automate the computation of word error rates (WER), identifying that humans significantly outperform models on isolated words, suggesting potential limitations in models' training data or acoustic context requirements
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PROJECTS

Security Analysis of Apple Intelligence

La Jolla, CA

Oct 2025 – Dec 2025

- Reverse-engineered macOS binaries using Ghidra to reconstruct the information flow between user input, Generative AI models, and local storage
- Identified a privacy vulnerability where the system persisted unencrypted sensitive personal information to local directories, bypassing the ephemeral-only privacy policy
- Developed a Python forensic script to simulate TCC (Transparency, Consent, and Control) inheritance attacks, successfully exfiltrating sensitive user data from sandboxed environments

User Intent Prediction on Shopping Sites

La Jolla, CA

Oct 2025 – Dec 2025

- Developed a dual Hidden Markov Model (HMM) system with the hmmlearn library to predict user purchase intent from clickstream data, achieving 90.4% accuracy and 0.92 precision on the RetailRocket dataset
- Engineered a high-volume data pipeline to process 2.7 million behavioral events, implementing a dynamic algorithm to reconstruct user browsing sessions
- Implemented the Baum-Welch algorithm for unsupervised training and the Forward algorithm for inference, classifying user sessions by comparing log-likelihood ratios between the two HMMs

Multi-Agent System for Science

La Jolla, CA

Apr 2025 – Aug 2025

- Managed a Kubernetes-based deployment pipeline to orchestrate multi-LLM-agent workloads across tasks on math problem-solving and complex conversations
 - Profiled computational bottlenecks and memory usage across agent interactions using NVIDIA Nsight Systems, identifying key inefficiencies in concurrent agent execution
 - Conducted system-level performance analysis to support research targeting architecture conferences
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SKILLS

- **Programming Languages:** Python, Java, C, C++, JavaScript, HTML, CSS, SQL, ARM, Clojure
- **Libraries & Tools:** Pandas, NumPy, PyTorch, Scikit-learn, Git, Kubernetes, Docker, Ghidra
- **Concepts:** Data Structures, Algorithms, Machine Learning, Data Analytics, Agile, Operating Systems, Computer Architecture, Database Systems, Cryptography, Computer Vision, Computational Linguistics
- **Languages:** Native/bilingual proficiency for Cantonese, English, and Mandarin; intermediate in Japanese